Vito **Kortbeek**

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Profile

Compiler engineer with hands-on experience across the complete LLVM/Clang pipeline, with most of my work focused on the middle end. I combine broad knowledge of processor architectures with experience across the front end, analyses and transformations, code generation, custom backends, and optimizations tailored to specific hardware features. I am particularly interested in alias and memory-dependence analysis and transformations that unlock performance by restructuring code. These interests also shaped my PhD research.

Experience

GlobalFoundries — *Remote*

MEMBER OF TECHNICAL STAFF, SOFTWARE ENGINEERING — *Jun. 2026 - present*

- Continuing development of ASIP Designer's extensively customized LLVM/Clang-based C/C++ compiler after the ASIP business transferred from Synopsys to GlobalFoundries in June 2026.

Synopsys — *Remote*

STAFF SOFTWARE ENGINEER — *Jan. 2023 - Jun. 2026*

- Developed ASIP Designer's extensively customized LLVM/Clang-based compiler, a mature product codebase with more than a decade of product-specific development on top of upstream LLVM. ASIP Designer generates a C/C++ SDK and synthesizable RTL for an application-specific processor from its architecture description.
- Worked across the full compiler pipeline, including the Clang front end, LLVM middle end, and a customized backend that emits a proprietary format for downstream ASIP tools. Most development focused on middle-end analyses, transformations, and optimization-pipeline support.
- Also contributed to separate proprietary compiler infrastructure and backend tooling built around compilation models distinct from LLVM/GCC.
- Developed product-specific code-restructuring optimizations and identified further opportunities by profiling workloads, inspecting generated assembly, and relating bottlenecks to ASIP hardware requirements.
- Unlike a conventional fixed-target compiler, this compiler dynamically supports processor targets derived from architecture descriptions, ranging from RISC-V-based processors to complex, highly specialized DSP architectures.
- Contributed code and RFC work to ongoing LLVM upstream support for C `restrict`-qualified local pointer variables, extending the existing function-parameter support; co-led a roundtable on the proposal at an LLVM conference.
- Received two Synopsys Certificates of Recognition for Outstanding Individual Contribution, awarded in Q4 FY23 and Q2 FY25.

Delft University of Technology — *Delft, the Netherlands*

PHD CANDIDATE — *Jan. 2019 - Dec. 2022*

- Researched programmer-friendly intermittent computing, with work spanning compiler support, architecture, and user-facing systems.
- Published at PLDI, ASPLOS, and UbiComp/IMWUT.

Micro Elektronische Producten (MEP) — *Landsmeer, the Netherlands*

JUNIOR SOFTWARE DEVELOPER — *Feb. 2014 - Sep. 2015*

- Developed software for an in-house Texas Instruments DSP platform running an RTOS for Voice over IP applications, including SNMP support and a zero-copy messaging scheme between tasks.

Education

Delft University of Technology — *Delft, the Netherlands*

PHD · DISSERTATION: ACCELERATING PROGRAMMER-FRIENDLY INTERMITTENT COMPUTING · DEFENDED 29 JUN. 2023 — *Jan. 2019 - Jun. 2023*

Delft University of Technology — *Delft, the Netherlands*

MSC EMBEDDED SYSTEMS (+ ONE-YEAR PRE-MASTER) — *Sep. 2015 - Feb. 2019*

Amsterdam University of Applied Sciences — *Amsterdam, the Netherlands*

BACHELOR ELECTRICAL ENGINEERING (HBO) — *Sep. 2011 - Jul. 2015*

Publications

Data Cache for Intermittent Computing Systems with Non-Volatile Main Memory **ASPLOS '25**
ACM International Conference on Architectural Support for Programming Languages and Operating Systems *Mar. 30 - Apr. 3, 2025*
Sourav Mohapatra, **Vito Kortbeek**, Marco Antonio van Eerden, Jochem Broekhoff, Saad Ahmed, Przemysław Pawełczak

NACHO integrates write-after-read detection into a data cache and reduced runtime overhead by 54% on average.
https://doi.org/10.1145/3676641.3715989
https://github.com/TUDSSL/intermittent-risc-v/tree/nacho-artifact-release

WARio: Efficient Code Generation for Intermittent Computing **PLDI '22**
ACM SIGPLAN Conference on Programming Language Design and Implementation *Jun. 13 - 17, 2022*
Vito Kortbeek, Souradip Ghosh, Josiah Hester, Simone Campanoni, Przemysław Pawełczak

Compiler transformations that reduced checkpoint overhead by 58% on average and up to 88%.
https://doi.org/10.1145/3519939.3523454
https://github.com/TUDSSL/WARio

BFree: Enabling Battery-free Sensor Prototyping with Python **UbiComp '21**
ACM International Joint Conference on Pervasive and Ubiquitous Computing *Sep. 21 - 26, 2021*
Vito Kortbeek, Abu Bakar, Stefany Cruz, Kasim Sinan Yildirim, Przemysław Pawełczak, Josiah Hester

A Python-based platform for prototyping battery-free sensors.
https://doi.org/10.1145/3432191
https://github.com/TUDSSL/BFree
Adafruit, Hackaday, The Independent, de Volkskrant

Battery-Free Game Boy **UbiComp '20**
ACM International Joint Conference on Pervasive and Ubiquitous Computing *Sep. 12 - 17, 2020*
Jasper de Winkel, **Vito Kortbeek**, Josiah Hester, Przemysław Pawełczak

An energy-harvesting handheld gaming platform that operates without batteries.
https://doi.org/10.1145/3411839 ***Distinguished Paper Award***
https://github.com/TUDSSL/ENGAGE
The Wall Street Journal, The Verge, CNET, Engadget

Time-sensitive Intermittent Computing Meets Legacy Software **ASPLOS '20**
ACM International Conference on Architectural Support for Programming Languages and Operating Systems *Mar. 16 - 20, 2020*
Vito Kortbeek, Kasim Sinan Yildirim, Abu Bakar, Jacob Sorber, Josiah Hester, Przemysław Pawełczak

Compiler and runtime support for running legacy embedded software intermittently.
https://doi.org/10.1145/3373376.3378476
https://github.com/TUDSSL/TICS

Earlier Projects

Head Communications, ESA Business Incubation Centre *Noordwijk, the Netherlands*
Wireless Sensor Systems Intern (BSc graduation project) *Feb. 2015 - Sep. 2015*
GPS, GSM, and Bluetooth LE connectivity; Embedded Linux; ZeroMQ-based data distribution; Android tooling.

HvA CleanMobility: Shell Eco-marathon and Solar Boat Challenge *Amsterdam, the Netherlands*
Embedded Hardware and Software Developer (extracurricular project) *Sep. 2012 - Aug. 2013*
OLED display; power-failure-resilient data logging; GPS; brushed and brushless motor control; telemetry; first-place H2A vehicle.

Skills

Compiler development	LLVM/Clang middle end, memory-dependence analysis, transformations, retargetable compilers
Programming	C, C++, Python, Assembly, Bash, Java, Lua, VHDL
Architectures	RISC-V, ARM, MSP430, ATmega, many custom ASIP and DSP architectures
Tools	Git, CMake, Make, GDB, Docker, GNU/Linux, Embedded Linux, RTOS
Languages	Dutch (native), English (C1; IELTS 8)